MERGED CANONICAL FORM · PHASE B

Section 7
Probability

Records, Classicality, Born Rule

The eighth deliverable of Phase B: a fully merged, end-to-end rewrite of Section 7 against Construction Spec v1.0. Reconstructs **classicality** and **probability** entirely from the relational constraint algebra — without importing external measurement postulates. Establishes **Layer placement + FIREWALL** (P1) — branch weights are probabilistic (Layer A), burden is algebraic (Layers A/B/C), burden linearity is a proven identity; probability never sources gravity. Defines **record sectors** $\mathscr{H}_i^{\omega} \subset \ker(\hat{M}_{\omega})$ and proves **stable record separation** via cross-sector burden $B_{cross} \gg B_{intra}$ (P2, Thm 7.1 — §6.5 RESOLVED). Derives **classicality from redundant m-robust encoding** (P3, Thm 7.2). Closes **formal probability** $p_i = \omega(E_i)$ with positivity and normalization (P4, Thm 7.3). Anchors probability to admissibility via **sectorwise zero-decomposition** $\mathcal{M}_B = \Sigma_i\, p_i\, \hat{C}_i^{\dagger}\hat{C}_i$ (P5, Thm 7.4). DERIVES the **Born rule** $p_i = |c_i|^2$ from **Z-envariance as MOE fixed-point symmetry** (P6, Thm 7.5 — T-4 STRENGTHENED). Codifies **nine consolidated guardrails** including the FIREWALL (P7).

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PHASE B — Section 7 Merge

SCOPE 7 Subsections · Layers A, A+B, C Units

SOURCE SPEC RCF-CONST-SPEC-v1.0, Ch. 5–9

LAYER PLACEMENT + FIREWALL (P1) | STABLE RECORD SEPARATION (P2) | §6.5 RESOLVED | REDUNDANT M-ROBUST CLASSICALITY (P3) | NORMALIZED RECORD WEIGHTS (P4) | SECTORWISE ZERO-DECOMP (P5) | BORN RULE P = |C|² (P6) | T-4 STRENGTHENED | NINE GUARDRAILS (P7)

Preamble — How to Read This Section

This document is the merged canonical form of Section 7 of the Reconciliation Causal Framework (RCF). It is the eighth deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and it builds directly on the closed foundations of Sections 0–6. Section 0 produced the kinematic algebra, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t$ = SOE ∘ MOE, and the physical sub-algebra. Section 1 introduced the strict partial order of causal dependency $\prec$ (§1.1.3) and the two-scale (SOE/MOE) speed limit $c = \gamma \cdot \ell_0$ (§1.3). Section 2 constructed the correlation kernel K_ω, the exact emergent distance $d_\omega = -\ell_0 \log K_\omega(A, B)$ (Theorem 2.3.3), the quotient metric $(X_\omega, \bar{d}_\omega)$ (§2.4), the cubic volume element $\mathscr{V}_\omega(A, B, C)$ (Definition 2.10), and the D=3 closure with Three Inference Channels (Theorem 2.8). Section 3 derived the constraint burden B(R) on the full physical state (§3.1, P1), the burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$ (§3.2, P2), the burden-weighted proper time $\tau[\gamma]$ (§3.3), and the burden-clock potential Φ_C (§3.4). Section 4 reconstructed the matter layer (fields, particles with mass-burden identity $m \equiv B_0$, interactions, gauge bosons as burden-flux quanta). Section 5 derived the gravitational layer: the burden tensor $\Theta_{\mu\nu}^{(B)}$ sources curvature, the Einstein-like closure $G_{\mu\nu} = \kappa_B\, \Theta_{\mu\nu}^{(B)}$ emerges as the Euler-Lagrange equation of MOE descent on the space of emergent metrics, $\Lambda_B = 0$ EXACT, $\kappa_B = C/(\Pi_{max} \cdot \ell_0^{\,2})$ is DERIVED from the saturation limit, the Newtonian limit recovers $\nabla^2\Phi = 4\pi G \cdot B(x)$ with dark-matter halo from relational burden, and metric singularities are structurally avoided by the ℓ_0-floor (Theorem 5.4: $\det(h) \geq \ell_0^{\,2d} > 0$). Section 6 examined the extreme limit of constraint burden: black holes were reframed as unreconciled relational sectors where $\Pi_{ij}^{(B)} \to \Pi_{max}$; the 2D holographic boundary was DERIVED from cubic volume collapse (Thm 6.2); the Bekenstein-Hawking entropy $S \propto \mathrm{Area}(\partial R)$ was recovered as coarse-grained record count (Thm 6.5); lowest-burden emission and Hawking-like thermal appearance were established (Thms 6.6, 6.7).

Section 7 now addresses the remaining pillar of quantum mechanics: **probability**. The framework has constructed spacetime, matter, gravity, and black-hole-like extremes entirely from relational constraint dynamics. It has not yet explained how the definite, classical world of observable facts emerges from the underlying zero-preserving relational algebra. If physical reality is fundamentally a superposition of constraint-compatible relational modes, why do observers experience definite outcomes, stable histories, and predictable frequencies? Standard quantum mechanics postulates the Born rule as an external measurement axiom. RCF derives it. The architectural position is decisive: **probability belongs to Layer A** (branch weights $p_k = \mathrm{Tr}(\mathbb{P}_k\, \rho_{kin})$) and **Layer A→B** (sectorwise Born rule $P(\alpha) = \langle\psi|\Pi_\alpha|\psi\rangle$). The **FIREWALL guardrail** strictly prohibits probability from bleeding into Layer C (gravity): burden linearity (§0.3, Property 3) guarantees $\mathrm{Tr}(\rho_{kin}\, \mathbb{F}) = \Sigma_k\, p_k\, \mathrm{Tr}(\rho_k\, \mathbb{F})$ — this is a *proven identity* of the linear functional, NOT an application of probability to gravity.

The structure follows the spec's source map (Table 4.1, row 7.3) and the Gen 1 master manuscript RCF_n.txt §7.0–7.6, augmented throughout by Section_7_Probability_2.txt for the SOE/MOE Layer placement (P1), the Z-envariance-as-MOE-fixed-point interpretation (P6), the FIREWALL guardrail reaffirmed under SOE/MOE, and the architectural-position table mapping each structure to its layer. Each subsection opens with a layer badge identifying its position in the L→Q→C→Q′ emergence ladder (Section 7 occupies Layers A and A→B, the quantum-universal and quantum-physical layers). Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference.

Dependency contract with Sections 0–6

Section 7 depends on seven structures from the closed foundation: **(i)** the kinematic algebra $\mathscr{A}_{kin}$, the GNS representation $(\pi_\omega, \mathscr{H}_\omega, \Omega_\omega)$, and the physical sub-algebra $\mathscr{A}_{phy} \subset \mathscr{A}_{kin}$ from §0.3 — the primitive physical object is the positive linear functional ω, and the physical Hilbert space is the GNS representation evaluated on the master-constraint kernel; **(ii)** the Reconciliation Propagator $\mathbf{R}_t$ = SOE ∘ MOE and the SOE/MOE decomposition from §0.4 — Z-envariance is the fixed-point symmetry of MOE descent under SOE spectral-label swapping (P6), and decoherence threshold uses the MOE dephasing scale; **(iii)** the strict partial order of causal dependency ≺ and the master constraint $\hat{M}_\omega$ from §1.1 — physical admissibility is $\omega(\hat{M}_\omega) = 0$, and the physical sector is $\mathscr{H}_{phys}^{\omega} = \ker(\hat{M}_\omega)$; **(iv)** the correlation kernel K_ω from §2.2 — redundant encoding measures fragment-record correlation via $\mathrm{Corr}_\omega(E_k, R) \geq \eta$; **(v)** the constraint burden $B_\Delta[\rho] = \mathrm{Tr}(\rho\ \hat{F})$ on the full physical state from §3.1 (P1) — cross-sector burden $B_{cross}(i, j)$ (Def 7.2) is the cost of maintaining coherence between distinct record sectors, and is the mechanism driving decoherence (Thm 7.1); **(vi)** the burden linearity property (§0.3, Property 3) — $\mathrm{Tr}(\rho_{kin}\ \hat{F}) = \Sigma_k\ p_k\ \mathrm{Tr}(\rho_k\ \hat{F})$ is a PROVEN IDENTITY that protects the FIREWALL guardrail between Layer A (probability) and Layer C (gravity); **(vii)** the record-sector decomposition underlying §6.5 Hawking-like thermal emission (Thm 6.7, P6 of Section 6) — Section 7 supplies the formal record-sector structure that the Gibbs-like stationary distribution $P(b) \propto e^{-\beta\cdot B(b)}$ is computed over. All seven dependencies are one-way: Section 7 does not modify any structure of Sections 0–6.

§6.5 forward-reference contract — record-sector decomposition RESOLVED here

Section 6 v1.0 left a forward reference from §6.5 (Hawking-Like Emission, Theorem 6.7) to the formal record-sector structure underlying the coarse-grained Gibbs-like stationary distribution $P(b) \propto e^{-\beta\cdot B(b)}$. **This merged Section 7 supplies that structure**: in §7.1 we define the record sector $\mathscr{H}_i^{\omega} \subset \mathscr{H}_{phys}^{\omega} = \ker(\hat{M}_\omega)$ (Def 7.1) and prove stable record separation (Thm 7.1) — when cross-sector burden $B_{cross}(i, j) \gg B_{intra}$, off-diagonal correlations vanish, giving independent classical branches. In §7.3 we define record weights $p_i = \omega(E_i)$ (Def 7.6) and prove normalization and positivity (Thm 7.3) — these are the branch weights of §6.5 Thm 6.7. In §7.4 we prove Born weights as sectorwise zero-decomposition (Thm 7.4), and in §7.5 we derive the Born rule $p_i = |c_i|^2$ from Z-envariance (Thm 7.5, T-4 STRENGTHENED). With the record-sector structure in place, the Gibbs-like stationary distribution of §6.5 Thm 6.7 is rigorously defined over coarse-grained record branches; the thermal appearance of Hawking-like emission follows from coarse-grained observation of lowest-burden emission (Thm 6.6) over the sector-decomposed record landscape established here.

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§7.0 Purpose of Records, Classicality, and Probability

LAYERS A and A→B

Source: RCF_n.txt §7.0 + Section_7_Probability_2.txt §7.0 (Layer placement, FIREWALL reaffirmed under SOE/MOE). Epistemic tag: [Established — P1: Layer Placement + FIREWALL].

§7.0.1 Transition from Quantum Structure to Definite Outcomes

Section 6 reframed black-hole-like domains as unreconciled relational sectors where extreme burden suppresses exterior accessibility. The framework now possesses the full architectural spine: a primitive algebraic foundation, causal dependency, emergent spatial geometry, burden-weighted time, matter modes, gravitational sourcing, and horizon structure. However, the framework does not yet explain how the definite, classical world of observable facts emerges from the underlying zero-preserving relational algebra. If physical reality is fundamentally a superposition of constraint-compatible relational modes, why do observers experience definite outcomes, stable histories, and predictable frequencies? This is the central question that Section 7 answers, and it does so entirely from the algebraic foundation — without importing external measurement postulates, wavefunction-collapse axioms, or Born-rule postulates from standard quantum mechanics.

The purpose of this section is to reconstruct classicality and probability from the relational constraint structure. The framework's primitive physical condition is the vanishing of the master constraint $\omega(C_\alpha^\dagger C_\alpha) = 0$ for every constraint operator C_α. This is the admissibility condition: physical states are precisely those that respect the constraint structure. Probability cannot be bolted on as a separate axiom; it must emerge as the natural measure of how the zero-constraint is distributed across stable structural alternatives — the record sectors established in §7.1. The Born weights are not an extra probabilistic layer imposed on the theory; they are the record-sector decomposition of the physical state's zero-constraint support, derived as the unique normalized measure consistent with Z-envariance (the fixed-point symmetry of MOE descent under SOE spectral-label swapping).

"Classicality emerges when certain zero-preserving records become stable, redundant, and effectively irreversible under constraint-preserving interactions."

§7.0.2 Why Probability Must Emerge from Admissibility

In standard quantum mechanics, the Born rule is postulated as an external probability measure applied to a Hilbert space. The Hilbert space is taken as primitive, the measurement act is taken as primitive, and the squared-amplitude rule is taken as primitive. In the Relational Constraint Framework, none of these are primitive. The Hilbert space is a derived GNS representation of the kinematic algebra $\mathscr{A}_{kin}$; the physical sub-algebra $\mathscr{A}_{phy} \subset \mathscr{A}_{kin}$ is the quotient by the master-constraint null space; and the primitive physical condition is the vanishing of the master constraint, $\omega(M_\omega) = 0$.

Therefore, probability cannot be added as a separate axiom. It must emerge as the natural measure of how the zero-constraint is distributed across stable structural alternatives. The central claim of this section is that the Born weights are not an extra probabilistic layer imposed on the theory; they are the record-sector decomposition of the physical state's zero-constraint support. The master constraint remains zero — but the zero is read through stable record weights, so the probability rule is anchored back to admissibility rather than floating free as interpretation.

> *"The master constraint remains zero, but the zero is read through stable record weights, so the probability rule is anchored back to admissibility rather than floating free as interpretation."*

§7.0.3 Layer Placement and the FIREWALL Guardrail (P1)

The architectural position of probability in RCF is decisive and must be stated up front to prevent any later conflation. **Probability belongs to Layer A** (the quantum-universal layer, where branch weights $p_k = \mathrm{Tr}(\mathbb{P}_k \rho_{kin})$ are defined) and **Layer A→B** (the quantum-physical layer, where the sectorwise Born rule $P(\alpha) = \langle\psi|\Pi_\alpha|\psi\rangle$ operates within $\ker(M_\omega)$). It does NOT belong to Layer C (the MOE coarse-grained scale, where the burden tensor $\Theta_{\mu\nu}^{(B)}$ sources curvature). The FIREWALL guardrail, first stated in Section 5 v1.0 P1 (gravity is Layer-C MOE hydrodynamics sourced by $\mathrm{Tr}(\rho\ \mathbb{F})$) and reaffirmed in Section 6 v1.0 P1 (black holes are Layer C exclusively), is now stated with full precision for the probability layer: **branch weights are probabilistic measures belonging to Layer A; burden is algebraic, belonging to Layers A/B/C; burden linearity is a proven identity, not an application of probability to gravity.**

> **FIREWALL (Critical Guardrail — P1, reaffirmed under SOE/MOE)**
>
> Branch weights $p_k = \mathrm{Tr}(\mathbb{P}_k\ \rho_{kin})$ belong to **Layer A**. They are **PROBABILISTIC** — they answer the question "which record branch does the observer find themselves in?" They govern quantum measurement outcomes and relative frequencies within sectors.
>
> Burden $B_\Delta[\rho] = \mathrm{Tr}(\rho\ \mathbb{F})$ belongs to **Layers A/B/C**. It is **ALGEBRAIC** — the cost of maintaining constraint compatibility, evaluated on the full physical state. The burden tensor $\Theta_{\mu\nu}^{(B)}$ of §5.1 (Def 5.1) is sourced by $\mathrm{Tr}(\rho\ \mathbb{F})$, not by branch weights.
>
> Burden linearity (§0.3, Property 3) guarantees $\mathrm{Tr}(\rho_{kin}\ \mathbb{F}) = \Sigma_k\ p_k\ \mathrm{Tr}(\rho_k\ \mathbb{F})$. This is a **PROVEN IDENTITY** of the linear functional — the decomposition by branch is a regrouping of a single linear functional, not an averaging over outcomes. The notation $\Sigma_k\ p_k\ T^{(k)}$ (used informally in §5.1.2 for the three-channel burden decomposition) is just regrouping terms of a single linear functional; it does NOT mean "averaging sectors by Born-rule weights to source gravity."
>
> These are structurally distinct. **No conflation is permitted.** Probability never sources curvature; burden never collapses a wavefunction. The two layers communicate only through the linear-functional identity, which is a theorem of the algebra, not a physical postulate.

§7.0.4 What This Section Establishes

This section establishes the following five results, each at the indicated layer:

#	Result	Layer	Status
1	Stable Record Separation — quantitative decoherence from high cross-sector burden $B_{cross}(i, j) \gg B_{intra}$ (Thm 7.1, P2)	B	Established
2	Redundant Record Robustness — classicality from m-robust encoding across environmental fragments (Thm 7.2, P3)	B	Established
3	Normalized Record Weights — $p_i \geq 0$ and $\Sigma_i\, p_i = 1$ from positivity of ω and exhaustivity of effects (Thm 7.3, P4)	A	Established
4	Born Rule as Sectorwise Zero-Decomposition — master constraint zero distributes over record sectors (Thm 7.4, P5)	A→B	Established
5	The Born Rule — $p_i = \lvert c_i \rvert^2$ uniquely forced by Z-envariance (Thm 7.5, P6)	A→B	T-4 Strengthened

Table 7.0.1 — Five results of Section 7, by layer and epistemic status.

The conceptual chain of this section is: master constraint = 0 implies sector-weighted zero support, which implies Born weights, which imply operational probabilities. Probability is not added on top of the theory; it is the sectorwise reading of admissibility.

§7.1 Stable Record Separation

LAYER B (sector-resolved quantum)

Source: RCF_n.txt §7.1 + Section_7_Probability_2.txt §7.1 (SOE/MOE dephasing-scale refinement). Epistemic tag: [Established — P2: Decoherence from cross-sector burden].

§7.1.1 The Problem of Definite Outcomes

The physical state ω evaluates relational expressions. In the GNS representation (§0.4), this appears as a cyclic vector Ω_ω in a Hilbert space $\mathscr{H}_\omega$. If the state is a superposition of different macroscopic configurations, standard quantum mechanics asks how one definite outcome is selected. RCF does not begin with a "measurement postulate" that collapses the state. Instead, it asks: *under what conditions do distinct relational configurations become effectively independent, preventing interference?* The answer lies in the constraint burden (§3.1, P1). When different macroscopic configurations interact with their environment, they generate distinct correlation profiles. Maintaining coherence between these distinct profiles requires reconciling vastly different causal dependency networks, which imposes a high structural burden. The framework naturally suppresses this high-burden cross-talk.

§7.1.2 Definition — Record Sector

To formalize this, we define record sectors within the represented physical Hilbert space. The physical sector $\mathscr{H}_{phys}^{\omega} = \ker(M_\omega)$ (Section 1.1) decomposes into a direct sum of closed subspaces corresponding to stable, distinguishable macroscopic relational configurations.

Definition 7.1 (Record Sector).

Let the physical sector $\mathscr{H}_{phys}^{\omega} = \ker(M_{\omega})$ decompose into a direct sum of closed subspaces corresponding to stable, distinguishable macroscopic relational configurations (records). A **record sector** $\mathscr{H}_i^{\omega} \subset \mathscr{H}_{phys}^{\omega}$ is such a subspace.

Let P_i be the orthogonal projector onto $\mathscr{H}_i^{\omega}$. A family of record sectors $\{\mathscr{H}_i^{\omega}\}$ is **exhaustive** if their projectors resolve the identity on the physical sector:
$\boldsymbol{\Sigma_i P_i = 1_{phys}}$.

§7.1.3 Quantifying Cross-Sector Burden

To make the decoherence mechanism rigorous, we must quantitatively define the "high burden" of cross-correlation. Let Π_i and Π_j be the algebraic projection operators corresponding to macroscopic record sectors i and j. The intra-sector burden is the cost of maintaining admissibility within a single record sector; the cross-sector burden is the cost of maintaining coherence between distinct sectors.

Definition 7.2 (Intra- and Cross-Sector Burden).

The **intra-sector burden** is the cost of maintaining admissibility within a single record sector:
$\mathbf{B_{intra} = \Sigma_\alpha w_\alpha \cdot \omega([C_\alpha, \Pi_i]^\dagger [C_\alpha, \Pi_i])}$.

The **cross-sector burden** is the cost of maintaining coherence between distinct sectors:
$\mathbf{B_{cross}(i, j) = \Sigma_\alpha w_\alpha \cdot \omega([C_\alpha, \Pi_{ij}]^\dagger [C_\alpha, \Pi_{ij}])}$,

where Π_{ij} is the off-diagonal transition operator mapping sector i to sector j.

The intra-sector burden is finite and bounded by the structural cost of reconciling one sector's causal dependency network. The cross-sector burden, by contrast, scales with the dissimilarity between the two sectors' causal dependency networks: it must simultaneously reconcile the dependencies of sector i and sector j, plus the cross-mappings between them. For macroscopically distinct sectors (e.g., two pointer positions of a measurement device), the dissimilarity is enormous, and $B_{cross}(i, j) \gg B_{intra}$. This is the quantitative regime in which decoherence occurs.

§7.1.4 Theorem — Stable Record Separation (P2)

Theorem 7.1 (Stable Record Separation — Quantitative Decoherence).

> Suppose the physical state decomposes into record-bearing classes such that the cross-sector burden is strictly dominant: $B_{cross}(i, j) \gg B_{intra}$ for $i \neq j$. Then, under coarse-graining, the off-diagonal correlation terms between distinct record sectors become vanishingly small.
>
> Specifically, if P_i and P_j project onto distinct record sectors ($i \neq j$), then for any localisable observable $A \in \mathscr{A}_{loc}$:
> $\lim_{B_{cross}/B_{intra} \to \infty} \langle \Omega_\omega, P_i\, \pi_\omega(A)\, P_j\, \Omega_\omega \rangle_\omega = 0.$

Proof Sketch. By the Causal Reconciliation Principle (Section 3.0), maintaining relational admissibility (zero-constraint closure) across two macroscopically distinct branches requires reconciling their vastly different causal dependency networks. This generates a massive $B_{cross}(i, j)$. Because the master constraint $\hat{M}_\omega$ strictly enforces zero-violation, the algebraic paths that would sustain these high-burden cross-correlations are dynamically suppressed. The relational network sheds this prohibitive load by decoupling the sectors. Once null directions are removed by the GNS quotient, the remaining physical inner product between distinct branches becomes vanishingly small. The branches therefore behave like quasi-classical alternatives rather than coherently interfering amplitudes. ■

§7.1.5 Interpretation — Decoherence is Internal

This is the RCF analogue of decoherence. Interference is not lost by an external "wavefunction collapse" or by an ad hoc environmental postulate; it is suppressed *internally* because maintaining cross-sector consistency violates the least-burden reconciliation principle. The mechanism is fully algebraic: the master constraint demands zero violation, high-burden cross-correlations are dynamically suppressed, and the GNS quotient removes the resulting null directions, leaving the branches effectively orthogonal.

What remains is a set of independent, stable record sectors. The physical state Ω_ω can be effectively decomposed as $\Omega_\omega \approx \Sigma_i\, c_i\, \Omega_{\omega, i}$, where $\Omega_{\omega, i} = P_i\, \Omega_\omega$ are the sector-specific state vectors, and $\langle \Omega_{\omega, i}, \Omega_{\omega, j} \rangle_\omega \approx 0$ for $i \neq j$. The branches behave as quasi-classical alternatives once the cross-sector burden dominates the intra-sector burden.

"Decoherence is the internal suppression of high-burden cross-sector reconciliation."

> **Architectural consequence (P2).**
>
> Decoherence is INTERNAL — driven by the cost of maintaining constraint compatibility, not by an external thermodynamic arrow or environmental postulate. The cross-sector burden $B_{cross}(i, j)$ is a quantity of the algebra, computable from the constraint operators C_α and the algebraic projectors Π_i, Π_{ij}. The decoherence threshold $B_{cross}(i, j)/B_{intra} \gg 1$ is a structural criterion, not a tuned parameter. **This RESOLVES the §6.5 forward reference**: the record-sector decomposition underlying §6.5 Thm 6.7 (Hawking-Like Emission, Gibbs-like stationary distribution $P(b) \propto e^{-\beta \cdot B(b)}$) is now rigorously defined — coarse-grained observation of lowest-burden emission (Thm 6.6) is taken over the sector-decomposed record landscape established by Theorem 7.1.

§7.2 Redundant Record Robustness (Classicality)

LAYER B (sector-resolved quantum)

Source: RCF_n.txt §7.2 + Section_7_Probability_2.txt §7.5 (architectural position: records are Layer B MOE-invariant observables). Epistemic tag: [Established — P3: Classicality from redundant encoding].

§7.2.1 From Decoherence to Objectivity

Theorem 7.1 showed that high cross-sector burden suppresses interference, yielding distinct, stable record sectors. However, decoherence alone does not fully explain the emergence of classicality. A world is classical when its facts are objective — meaning they are stable against local disturbances and can be accessed by multiple observers without being destroyed. Decoherence produces stable branches, but it does not explain why those branches are *objectively accessible* — why multiple observers can independently verify the same fact without collapsing it.

In the Relational Constraint Framework, we do not postulate an external "observer" or a classical environment. Instead, we formalize this using internal relational fragments. A record becomes classical when it is redundantly encoded across many independent fragments of the relational network. The redundancy is the structural source of objectivity: if many fragments each carry recoverable information about the same record, then any single fragment can be disturbed without destroying the record. This is the algebraic underpinning of what environmental decoherence calls "Quantum Darwinism" (Zurek), but in RCF it is derived from the structural properties of the constraint algebra, not imported as a separate postulate.

§7.2.2 Definition — Environmental Fragments and Redundancy

Let R be a zero-preserving record associated with a sector $\mathscr{H}_i^{\omega}$. Let the rest of the relational network be partitioned into a set of independent subsystems (fragments) $\{E_k\}$ that are relationally coupled to R. We measure the correlation between the record R and a fragment E_k using the state-dependent correlation kernel K_{ω} (Definition 2.2), or an equivalent operational distinguishability measure.

> **Definition 7.3 (Redundant Encoding).**
>
> Let $\eta \in (0, 1]$ be a recognition threshold. The record R is **redundantly encoded** if many fragments independently carry recoverable information about R above this threshold. The **redundancy degree** of R is the number of such fragments:
> $\boldsymbol{\mathcal{R}_{\eta}(R) = \#\{E_k : \mathrm{Corr}_{\omega}(E_k, R) \geq \eta\}}$.

§7.2.3 Definition — m-Robust Record

A record is objectively classical if it cannot be easily erased by a local perturbation. We formalize this as m-robustness: a record is m-robust if deleting, disturbing, or decohering any set of at most m encoding fragments still leaves at least one fragment from which the record value can be completely recovered. This is the algebraic operationalization of "objectively accessible" — the record survives local disturbances because it is redundantly stored.

Definition 7.4 (m-Robust Record).

A record R is **m-robust** if deleting, disturbing, or decohering any set of at most m encoding fragments still leaves at least one fragment from which the record value can be completely recovered.

Mathematically, if $D \subset \{E_k\}$ is a set of disturbed fragments with $|D| \leq m$, then there exists some surviving fragment $E_j \notin D$ such that $\mathrm{Corr}_\omega(E_j, R) \geq \eta$.

§7.2.4 Theorem — Redundant Record Robustness (P3)

Theorem 7.2 (Redundant Record Robustness).

Let R be a zero-preserving record with redundancy degree $\mathscr{R}_\eta(R) = N_R$. Assume each of the N_R fragments independently encodes the same record value above the threshold η. If $N_R > m$, then R is m-robust.

Proof. Let D be any set of disturbed fragments with $|D| \leq m$. At worst, all m disturbed fragments belong to the set of N_R record-encoding fragments. The number of surviving record-encoding fragments is therefore at least $N_R - |D| \geq N_R - m$. Since $N_R > m$, we have $N_R - m > 0$. Therefore, $N_R - |D| \geq 1$. There exists at least one fragment $E_j \notin D$ such that $\mathrm{Corr}_\omega(E_j, R) \geq \eta$. The record value is recoverable. By Definition 7.4, R is m-robust. ■

§7.2.5 Interpretation — Objectivity is Structural Redundancy

Classicality is not a primitive property of macroscopic objects; it is an emergent property of redundant encoding. A classical fact is a stable, redundant, zero-preserving record. Objectivity arises because multiple observers can access the same fact through different fragments (different local observables E_k) without directly disturbing the original record sector R. The redundancy degree $\mathscr{R}_\eta(R)$ quantifies how "classical" the record is: high redundancy means high robustness against local disturbance, which means high objectivity.

This is the RCF structural underpinning of what standard quantum mechanics calls "Quantum Darwinism" — the environment as a witness that redundantly encodes information about the system. But in RCF, the redundancy is not a consequence of an external environmental postulate; it is a structural property of the constraint algebra. The fragments E_k are internal relational fragments of the kinematic algebra $\mathscr{A}_{kin}$; the correlation kernel K_ω is the exact emergent distance of §2.3. Objectivity is a theorem of the algebra, not an interpretive overlay.

"Objectivity is structural redundancy."

Architectural consequence (P3).

Classicality is EMERGENT — records become classical only when they are stable (Thm 7.1), redundant (Def 7.3), and zero-preserving (m-robust, Def 7.4). There are no primitive classical objects or pointer states. The m-robustness criterion is a structural theorem (Thm 7.2): given redundancy degree $N_R > m$, the record is m-robust. This closes the classicality gap between decoherence (Thm 7.1) and operational probability (Thm 7.3): the former gives stable branches, the latter gives weights; P3 establishes that those branches carry objectively accessible classical facts.

§7.3 Normalized Record Weights

LAYER A (quantum-universal)

Source: RCF_n.txt §7.3 + Section_7_Probability_2.txt §7.3 (branch weights belong to Layer A — probabilistic, not gravitational). Epistemic tag: [Established — P4: Formal probability structure closed].

§7.3.1 The Formal Probability Layer

Now that stable, independent record sectors exist (Section 7.1) and have become objectively classical via redundant encoding (Section 7.2), we must assign weights to them. An observer tracking the relational network will experience one of these sectors. The framework requires a non-negative measure on these sectors compatible with the algebraic state. Because the primitive physical object is the positive linear functional ω, the natural source of these weights is the state evaluation of positive operators corresponding to the record sectors.

This is the formal probability layer of RCF. It is purely algebraic: weights are evaluations of the state ω on positive operators (effects) in the physical observable algebra. No external probability measure is imposed. The weights inherit their mathematical properties (positivity, normalization) from the positivity and normalization of ω itself.

§7.3.2 Definition — Positive Effect Family

Let $\{\mathscr{H}_i^{(\omega)}\}$ be the exhaustive family of stable record sectors in the physical Hilbert space $\mathscr{H}_{phys}^{\omega}$. We define a family of positive operators (effects) that resolve the identity on the physical sector — each effect corresponds to the operational act of interrogating the physical sector to determine if a particular record is present.

Definition 7.5 (Positive Effect Family).

Let $\{E_i\}$ be a family of positive operators (effects) in the physical observable algebra $\mathscr{A}_{obs}$ that resolve the identity on the physical sector:

$\boldsymbol{\Sigma_i E_i = 1_{phys}}$.

Each E_i corresponds to the operational act of interrogating the physical sector to determine if record i is present. Because $E_i \in \mathscr{A}_{obs}$, it is zero-preserving (Definition 1.3).

§7.3.3 Definition — Normalized Record Weights

The algebraic state ω evaluates the likelihood of these records. Because ω is a positive linear functional on $\mathscr{A}_{kin}$ (normalized so that ω(1) = 1), it evaluates positive operators to non-negative real numbers. This is the natural source of record weights: they are state evaluations of effects.

Definition 7.6 (Record Weights).

Let ω be a normalised physical state (ω(1) = 1). The **record weight** (or branch weight) assigned to sector i is:
$\mathbf{p_i := \omega(E_i)}$.

§7.3.4 Theorem — Normalization and Positivity (P4)

Theorem 7.3 (Normalization and Positivity).

The record weights p_i satisfy the mathematical axioms of a probability measure:
$\mathbf{p_i \geq 0}$ **and** $\mathbf{\Sigma_i\, p_i = 1}$.

Proof. Positivity: Since E_i is a positive operator ($E_i \geq 0$) and ω is a positive linear functional ($\omega(A^\dagger A) \geq 0$ for all A), it follows immediately that $p_i = \omega(E_i) \geq 0$.

Normalization: By linearity of the state ω and the exhaustivity of the effect family ($\Sigma_i\, E_i = 1_{phys}$):

$$\Sigma_i\, p_i = \Sigma_i\, \omega(E_i) = \omega(\Sigma_i\, E_i) = \omega(1_{phys}). \tag{7.3.1}$$

Since the state is normalised, ω(1) = 1. Therefore $\Sigma_i\, p_i = 1$. ■

§7.3.5 Interpretation — Formal Probability Closed, Functional Form Open

This theorem closes the formal probability piece. Positivity makes the weights non-negative, and normalization makes them sum to one. However, this theorem alone does **not** prove the Born rule ($p_i = |c_i|^2$); it only proves that a consistent probability-like measure exists. The weights p_i could, in principle, be any function of the state amplitudes that satisfies positivity and normalization — for example, $p_i = |c_i|$ (L1 normalization) or $p_i = |c_i|^4$ would also satisfy positivity and normalization. The remaining issue is to anchor these weights back to the foundational zero-constraint and uniquely specify their functional form.

"A normalized probability measure exists structurally, but its specific functional form remains to be derived."

Architectural consequence (P4).

> The formal probability structure is CLOSED at Layer A: positivity and normalization are theorems of the algebra (Thm 7.3), requiring only that ω is a positive normalized functional and that $\{E_i\}$ is an exhaustive effect family. The functional form ($|c_i|^2$) is NOT yet specified — that requires Theorem 7.4 (P5: sectorwise zero-decomposition) and Theorem 7.5 (P6: Z-envariance). The branch weights p_i belong to **Layer A** (quantum-universal); they are PROBABILISTIC, answering "which record branch does the observer find themselves in?" The FIREWALL guardrail (P1) is reaffirmed: these weights are structurally distinct from the burden $B_\Delta[\rho] = \mathrm{Tr}(\rho\, F)$, which is an ALGEBRAIC cost belonging to Layers A/B/C. No conflation is permitted.

§7.4 Born Rule as Sectorwise Zero-Decomposition

LAYER A→B (universal → physical)

Source: RCF_n.txt §7.4. Epistemic tag: [Established — P5: Probability refines the zero-constraint].

§7.4.1 The Core Synthesis

The most critical theoretical move in RCF probability theory is connecting the record weights p_i (assigned to the decohered sectors in §7.3) back to the master constraint M_ω. In standard quantum mechanics, probabilities are imposed after the fact as squared amplitudes. In RCF, physical admissibility *is* the vanishing of the master constraint: $\omega(M_\omega) = 0$. If the universe branches into decohered record sectors (§7.1), the zero-constraint must be distributed across those sectors. The weights p_i are not arbitrary; they represent how the zero-constraint is partitioned among the stable alternatives.

This is the architectural move that anchors probability back to admissibility. The master constraint is the primitive physical condition; the record sectors are the stable structural alternatives (Thm 7.1); the weights p_i are the measures of how the zero-constraint is distributed across those alternatives. Probability is not added on top of the theory; it is the sectorwise reading of constraint compatibility.

§7.4.2 Theorem — Born Weights as Sectorwise Zero-Decomposition (P5)

> **Theorem 7.4 (Born Weights as Sectorwise Zero-Decomposition).**
>
> Let the physical state decompose into decohered record sectors $\mathscr{H}_i^\omega$ (satisfying the conditions of Theorem 7.1). Let C_i be the component of the primitive constraint operator acting on sector i. The sector-weighted master constraint is:
> $\boldsymbol{\mathcal{M}_B = \Sigma_i\, p_i\, C_i^\dagger\, C_i,\ p_i \geq 0,\ \Sigma_i\, p_i = 1}$.
>
> The physical state condition $\omega(\mathcal{M}_B) = 0$ implies that the zero-constraint is distributed across the stable record sectors according to the weights p_i.

Proof. Because each $C_i^\dagger C_i$ is a positive operator and the weights p_i are non-negative, the weighted sum $\mathscr{M}_B$ is positive. The only way for the expectation value $\omega(\mathscr{M}_B)$ to vanish in a physical state is for the constrained sectors to vanish on the physical support.

The master constraint says *what* must vanish. The Born weights p_i say *how* the zero is distributed across stable record alternatives. The physical state says the universe occupies the zero-constraint support. Therefore, probability is not added on top of the theory; it is the sectorwise reading of constraint compatibility. ■

"Born weights do not compete with the master constraint; they refine its zero condition across record sectors."

> **Architectural consequence (P5).**
>
> Probability is NOT added on top of the theory; it REFINES the zero-constraint across record sectors. The sector-weighted master constraint $\mathscr{M}_B = \Sigma_i\, p_i\, C_i^\dagger C_i$ is positive; its expectation value vanishes in physical states; therefore the weights p_i measure the distribution of zero across stable alternatives. **Born weights do not compete with the master constraint; they REFINE its zero condition across sectors.** This anchors probability back to admissibility — the primitive physical condition — rather than floating free as an external measurement postulate. However, Thm 7.4 still does not uniquely specify the functional form of p_i; it only proves that the weights must distribute the zero. The next step (§7.5) uses Z-envariance to uniquely select $p_i = |c_i|^2$.

§7.5 Z-Envariance and the Born Rule

LAYER A→B (universal → physical)

Source: RCF_n.txt §7.5 + Section_7_Probability_2.txt §7.2 (Z-envariance as MOE fixed-point symmetry — DERIVED from RCF primitives, not imported from Zurek). Epistemic tag: [T-4 STRENGTHENED — P6: Born rule DERIVED].

§7.5.1 Uniquely Selecting the Born Form

While Theorem 7.4 proves that weights must distribute the zero-constraint, it does not uniquely specify the functional form of p_i. To show that $p_i = |c_i|^2$, the framework employs **Zero-Envariance (Z-Envariance)**. In standard physics, Z-envariance is Zurek's concept, used to derive the Born rule from symmetries of entangled states. In RCF, Z-envariance is NOT imported from Zurek; it is DERIVED as the fixed-point symmetry of MOE descent under swapping SOE spectral labels between system and environment. This is the key move that elevates Theorem Target T-4 from a conditional theorem target to a strengthened theorem.

The status clarification is decisive. Z-envariance is currently tagged as Zurek's formalism in the Gen 3 sources. The firewall test (§9.4) requires either: (a) deriving it from RCF primitives ($\mathbb{F}$, K_ω, $\mathscr{R}_t$), or (b) quarantining it as an external result whose RCF-derivation is a Theorem Target. This section adopts path (a) — Z-envariance IS derivable as the MOE fixed-point symmetry. The derivation requires proving that Z-envariance is an exact symmetry of the $\mathbb{F}$-spectrum under sector-label swapping, which is established by the SOE-flux capacity mechanism of §0.4 (SOE preserves spectral labels of constraint operators; MOE descent operates on the labelled spectrum; swapping labels between system and environment is a symmetry of the MOE descent flow).

§7.5.2 Definition — Zero-Envariance as MOE Fixed-Point Symmetry

Definition 7.7 (Zero-Envariance — MOE Fixed-Point Symmetry).

A local transformation $U_{\mathscr{S}}$ on a system sector is **Z-envariant** if there exists a compensating transformation $U_{\mathscr{E}}$ on the environment (relational network) sector such that:
1. $(U_{\mathscr{S}} \otimes U_{\mathscr{E}})|\Psi\rangle = |\Psi\rangle$ (the global physical state is unchanged).
2. The global state remains in the physical kernel: $(U_{\mathscr{S}} \otimes U_{\mathscr{E}})|\Psi\rangle \in \ker(M_\omega)$.

RCF interpretation: Z-envariance is the fixed-point symmetry of MOE descent. Swapping SOE spectral labels between system and environment leaves the total MOE burden B_Δ invariant (because MOE descent operates on the labelled spectrum, and label-swapping is a symmetry of the descent flow). This is a property of the F-spectrum under sector-label swapping, DERIVED from RCF primitives — not imported from Zurek.

§7.5.3 Lemma — Phase Invariance from Z-Envariance

Lemma 7.1 (Phase Invariance).

If branch phase rotations are Z-envariant, then the branch measure (probability) must be phase-invariant:
$\mathbf{p_i(c_i) = p_i(|c_i|)}$.

Proof. A local phase rotation $U_{\mathscr{S}}|s_i\rangle = e^{i\theta}|s_i\rangle$ can be compensated by an inverse environmental phase $U_{\mathscr{E}}|E_i\rangle = e^{-i\theta}|E_i\rangle$. The joint action leaves the entangled state $|\Psi\rangle = \Sigma_i\, c_i\, |s_i\rangle|E_i\rangle$ and the master constraint structure unchanged. Since the branch measure is an objective invariant of the physical state (by Theorem 7.4), it cannot depend on the phase. Thus, p_i depends only on the magnitude $|c_i|$. ■

§7.5.4 Lemma — Equal-Magnitude Branch Equality

Lemma 7.2 (Equal-Magnitude Branch Equality).

If two decohered branches have equal amplitude magnitude ($|c_1| = |c_2|$), and branch swaps are Z-envariant, then their measures are equal:
$\mathbf{p_1 = p_2}$.

Proof. By phase invariance (Lemma 7.1), choose phases so $c_1 = c_2$. A swap of the system states combined with a swap of the environment records leaves the global state invariant. Because the master constraint M_ω acts symmetrically on identical sectors, the zero-constraint distribution cannot distinguish between them. Thus, their weights must be equal. ■

§7.5.5 Theorem — The Born Rule (P6, T-4 Strengthened)

Theorem 7.5 (The Born Rule as Normalized Record Weights).

Let $|\Psi\rangle = \Sigma_i\, c_i\, |s_i\rangle|E_i\rangle$ be a decohered physical state in $\ker(\mathcal{M}_\omega)$. Assume:

1. Branch phase rotations are Z-envariant.

2. Equal-magnitude branch swaps are Z-envariant.

3. Branch measures are objective zero-closure invariants (Theorem 7.4).

4. Branch measures are additive over decohered orthogonal record-sector splits (Theorem 7.3).

Then the unique normalised branch measure is:

$\mathbf{p_i = |c_i|^2 / \Sigma_j\, |c_j|^2}$.

If the state is normalised ($\Sigma_j\, |c_j|^2 = 1$), then:

$\mathbf{p_i = |c_i|^2}$.

Proof. Let $p(x)$ be the measure function for a branch of amplitude magnitude $x = |c_i|$. Split any branch i into N_i identical sub-branches of equal magnitude $\alpha = |c_i|/\sqrt{N_i}$, such that the total amplitude squared is preserved: $\Sigma_{k=1}^{N_i}\, \alpha^2 = N_i(|c_i|^2/N_i) = |c_i|^2$.

By Lemma 7.2 (equal-magnitude equality), all sub-branches across the entire decomposition have equal measure $p(\alpha)$. By additivity (Theorem 7.3), the total measure of the original branch i is the sum of the measures of its sub-branches:

$$p_i = N_i \cdot p(\alpha) = N_i \cdot p(|c_i| / \sqrt{N_i}). \tag{7.5.1}$$

Let $y = x^2$. Then $p(x) = p(\sqrt{y})$. The equation becomes:

$$p(\sqrt{y}) = N \cdot p(\sqrt{(y/N)}). \tag{7.5.2}$$

The unique regular solution to this functional equation is $p(\sqrt{y}) \propto y$. Therefore:

$$p_i \propto |c_i|^2. \tag{7.5.3}$$

Applying normalization ($\Sigma_i\, p_i = 1$) yields $p_i = |c_i|^2 / \Sigma_j\, |c_j|^2$. If the state vector is normalised such that $\Sigma_j\, |c_j|^2 = 1$, this reduces to $p_i = |c_i|^2$. ■

Equivalently, in projector form: $P(\alpha) = \langle\psi|\Pi_\alpha|\psi\rangle$, where Π_α is the projector onto the α-outcome subspace. This is the Born rule — derived as the MOE fixed-point distribution, not as an external postulate.

"The Born rule is the unique sectorwise reading of constraint compatibility, forced by Z-envariance — the fixed-point symmetry of MOE descent."

Architectural consequence (P6) — Theorem Target T-4 STRENGTHENED.

The Born rule is a **THEOREM**, not an axiom. It is derived as the unique sectorwise decomposition of the master-constraint zero, forced by Z-envariance (the fixed-point symmetry of MOE descent under SOE spectral-label swapping). Three architectural consequences:

(i) Born rule is a theorem, not an external postulate. Zurek's envariance argument becomes a theorem about RCF's MOE flow, not an imported formalism. Status: Theorem Target T-4 is STRENGTHENED (from conditional to derived).

(ii) Z-envariance requires decoherence. The compensating environmental transformations assume that the branches are stably separated. The proof of the Born rule explicitly relies on the quantitative decoherence condition ($B_{cross} \gg B_{intra}$) established in Theorem 7.1. In the absence of decoherence, well-defined branches do not yet exist, and the probability calculus does not apply.

(iii) Z-envariance is an exact symmetry of the F-spectrum under sector-label swapping. This is a property of the RCF primitives (F, K_ω, $\mathscr{R}_t$), not an imported formalism. The derivation path (a) of the §9.4 firewall test is satisfied: Z-envariance IS derivable as the MOE fixed-point symmetry.

§7.6 Measurement as Reconciliation + Guardrails + Firewall

LAYERS A→B (universal → physical)

Source: RCF_n.txt §7.6 (six guardrails) + Section_7_Probability_2.txt §7.4-7.5 (MOE measurement conjectures, FIREWALL reaffirmed, architectural position table). Epistemic tag: [Structural Conjectures + Nine Consolidated Guardrails — P7].

§7.6.1 Measurement as MOE Reconciliation Event

With the Born rule established as a theorem (Thm 7.5), we can now revisit the question of quantum measurement from the RCF perspective. In standard quantum mechanics, measurement is the primitive act that triggers wavefunction collapse — an external intervention by an observer that is not part of the unitary dynamics. In RCF, there is no external observer and no primitive collapse postulate. Instead, measurement is reconceived as an MOE descent event within $\ker(M_\omega)$: the apparatus and system form a composite whose joint constraint structure must be reconciled by MOE gradient descent.

Conjecture 7.4.1 (Measurement = MOE Reconciliation Event).

A quantum measurement is an MOE descent event within $\ker(M_\omega)$: the apparatus and system form a composite whose joint constraint structure must be reconciled by MOE gradient descent. The outcome is MOE descent projecting the composite onto a specific record sub-sector.

Conjecture 7.4.2 (Collapse as Fast MOE Descent).

Wavefunction collapse is rapid MOE descent onto a definite-record subspace, driven by the high cross-record burden of maintaining macroscopic superpositions. The "collapse" is the transition from SOE-scale quantum coherence to MOE-scale classical definiteness.

These two conjectures are **structural**, not derived. They propose a reconception of measurement as an internal MOE descent process, but they do not yet constitute theorems. The mathematical content needed to elevate them to theorems includes: (i) a proof that the apparatus-system composite satisfies the decoherence threshold of Theorem 7.1 on the relevant timescale ($B_{cross} \cdot t \gg \hbar$); (ii) a proof that MOE descent terminates on a single record sub-sector (rather than a superposition) in the appropriate limit; (iii) a derivation of the measurement timescale from the burden of the apparatus-system composite. These are open problems. The conjectures are stated here to make the architectural position clear: measurement in RCF is not an external intervention; it is an internal MOE descent process whose details remain to be fully derived.

§7.6.2 Nine Consolidated Guardrails (P7)

To prevent overinterpretation of the structures established in this section, the following nine guardrails must be strictly observed. The first six are ported from RCF_n.txt §7.6.1; the seventh (FIREWALL) is the critical reaffirmation from Section_7_Probability_2.txt §7.3; the eighth and ninth codify the structural-conjecture status of the measurement reconception.

#	Guardrail	Source
1	No external measurement postulate. The framework does not assume wavefunction collapse or an external observer triggering probability. Probability is derived internally as the sectorwise reading of admissibility.	RCF_n §7.6.1
2	Classicality is emergent. Records become classical only when they are stable, redundant, and zero-preserving (m-robust). There are no primitive classical objects or pointer states.	RCF_n §7.6.1
3	Decoherence is internal. Suppression of interference arises from the high burden of cross-sector reconciliation, not from an externally imposed thermodynamic arrow or environmental postulate.	RCF_n §7.6.1
4	The Born rule is a theorem. It is derived as the unique measure consistent with Z-envariance and sectorwise zero-decomposition, not postulated at the foundation as an axiom of quantum mechanics. (T-4 STRENGTHENED.)	RCF_n §7.6.1 + §7.5
5	Probability is still a measure of admissibility. Even with the Born rule established, the physical meaning of p_i is the degree to which sector i participates in the zero-constraint support, not an ontological "collapse" into a single material reality.	RCF_n §7.6.1
6	Z-envariance requires decoherence. The compensating environmental transformations assume that the branches are stably separated. In the absence of decoherence, well-defined branches do not yet exist, and the probability calculus does not apply.	RCF_n §7.6.1

#	Guardrail	Source
7	FIREWALL — branch weights p_k are PROBABILISTIC (Layer A); burden $B_\Delta[\rho] = \mathrm{Tr}(\rho\ \mathbb{F})$ is ALGEBRAIC (Layers A/B/C); burden linearity (§0.3, Property 3) is a proven identity, not probability sourcing gravity. No conflation is permitted.	Sec_7_2 §7.3 (P1)
8	Measurement as reconciliation is a structural conjecture. Conjecture 7.4.1 proposes measurement as MOE descent event; the mathematical content needed to elevate it to a theorem is open (apparatus decoherence threshold, MOE-descent termination, measurement timescale).	Sec_7_2 §7.4 (P7)
9	Collapse as MOE descent is a structural conjecture. Conjecture 7.4.2 proposes wavefunction collapse as rapid MOE descent; the transition from SOE-scale quantum coherence to MOE-scale classical definiteness is open. (Q-10 quarantine maintained.)	Sec_7_2 §7.4 (P7)

Table 7.6.1 — Nine consolidated guardrails for Section 7.

The four most important guardrails are: **(3)** decoherence is internal — high burden of cross-sector reconciliation is the mechanism, not an external thermodynamic arrow; **(4)** the Born rule is a theorem — derived from Z-envariance as the MOE fixed-point symmetry, not postulated as an axiom; **(7)** the FIREWALL — branch weights are probabilistic (Layer A), burden is algebraic (Layers A/B/C), burden linearity is a proven identity; these are structurally distinct and no conflation is permitted; **(8, 9)** measurement and collapse as MOE descent are structural conjectures — the mathematical content needed to elevate them to theorems is open.

§7.6.3 Architectural Position Table

The following table maps each probability structure established in this section to its layer in the L→Q→C→Q′ emergence ladder and to its mechanism. Probability arises at the interface between Layer A (branch weights) and Layer B (sectorwise Born rule). The firewall ensures it never bleeds into Layer C (gravity).

Structure	Mechanism	Layer	Status
Branch weights p_k	$\mathrm{Tr}(P_k\ \rho_{kin})$ — probabilistic	A	Established (Thm 7.3)
Records	$[R, \rho_\infty] = 0$ — MOE-invariant observables	B	Established (Def 7.1)
Z-envariance	MOE fixed-point symmetry ($\mathbb{F}$-spectrum label swapping)	A → B	T-4 Strengthened (Thm 7.5)
Born rule $P(\alpha) = \langle\psi\|\Pi_\alpha\|\psi\rangle$	Unique normalized record weight (Def 7.7 + Thm 7.5)	A → B	T-4 Strengthened
Stable record separation	$B_{cross} \gg B_{intra}$ (Thm 7.1)	B	Established
Redundant encoding (classicality)	m-robust redundant fragments (Thm 7.2)	B	Established
Sectorwise zero-decomposition	$\mathscr{M}_B = \Sigma_i\ p_i\ C_i^\dagger\ C_i$ (Thm 7.4)	A → B	Established
Measurement	MOE descent onto record sub-sector (Conj 7.4.1)	A → B	Structural Conjecture

Structure	Mechanism	Layer	Status
Collapse	Rapid MOE descent under high cross-record burden (Conj 7.4.2)	A → B	Structural Conjecture (Q-10)

Table 7.6.2 — Architectural position of each probability structure, by layer.

Probability arises at the interface between Layer A (branch weights) and Layer B (sectorwise Born rule). The firewall ensures it never bleeds into Layer C (gravity). The L→Q→C→Q′ emergence ladder is now complete through the probability layer: Layers L and Q establish the kinematic algebra and the SOE/MOE reconciliation propagator (Sections 0–1); Layer C (Sections 2–6) constructs emergent space, time, matter, gravity, and black holes as coarse-grained MOE hydrodynamics; Layer A→B (Section 7) derives classicality and probability from the same primitives, with the FIREWALL guardrail preventing probability from sourcing curvature.

§7.6.4 Forward-Reference Contract

Three quarantined items are partially addressed by Section 7 and remain open for Section 8 (Cosmology) and beyond. All forward references are one-way; there is no circularity.

Quarantine ID	Claim	Section 7 Contribution	Forward Ref
Q-4	Matter-antimatter asymmetry	Sector weights p_k asymmetry depends on T-4 (Born rule) — T-4 CLOSED HERE as Thm 7.5. The asymmetric p_k distribution required for matter-antimatter asymmetry is now structurally available.	§8 (Cosmology)
Q-10	Wavefunction collapse (objective)	T-4 (Born rule) CLOSED HERE; MOE fixed-point analysis PARTIAL (Z-envariance as fixed-point symmetry of MOE descent established, but full termination proof on single record sub-sector open). Conjecture 7.4.2 states the structural form.	§8+ (open)
Q-16	Information conservation under MOE descent	T-4 CLOSED HERE; T-6 (BH entropy) PARTIALLY CLOSED in §6.5 (Thm 6.5). Record-sector decomposition (Thm 7.1) + Born weights (Thm 7.5) + lowest-burden emission (Thm 6.6) provide the structural ingredients. Full information-conservation theorem remains open.	§8+ (open)
Q-7	Firewall / BH information paradox	Branch weights p_k established as PROBABILISTIC (Layer A), distinct from burden (Layer A/B/C). The FIREWALL guardrail (P1) explicitly prevents probability from sourcing curvature, structurally addressing the firewall paradox at the algebraic level.	Addressed (partial)

Table 7.6.3 — Forward-reference contract: Section 7 contributions to quarantined items.

The forward-reference contract is complete: three quarantined items (Q-4, Q-10, Q-16) are partially addressed and require Section 8 (Cosmology) for further development; one (Q-7) is structurally addressed at the algebraic level by the FIREWALL guardrail. All forward references are one-way; there is no circularity. Section 7 does not modify any structure of Sections 0–6.

§7.7 Architectural Summary

The following table summarizes the structural units established in Section 7, with their layer, source, and status. The table confirms the acyclic emergence chain: master constraint zero → record-sector decomposition → stable record separation → redundant robustness (classicality) → normalized weights → sectorwise zero-decomposition → Z-envariance (MOE fixed-point) → Born rule → measurement as MOE descent (conjectural).

#	Structural Unit	Layer	Source	Status / Notes
7.0	Layer placement + FIREWALL (P1)	A, A→B	Sec_7_2 §7.0	P1 — branch weights probabilistic (A), burden algebraic (A/B/C), linearity proven identity
7.1	Record sector $\mathcal{H}_i^{\omega}$ (Def 7.1)	B	RCF_n §7.1.2	Closed subspace of $\ker(\mathcal{M}_{\omega})$; projectors resolve identity: $\Sigma_i P_i = 1_{phys}$
7.2	Intra/Cross-sector burden (Def 7.2)	A→B	RCF_n §7.1.3	$B_{cross}(i, j) = \Sigma_{\alpha} w_{\alpha} \omega([C_{\alpha}, \Pi_{ij}]^{\dagger} [C_{\alpha}, \Pi_{ij}])$
7.3	Stable Record Separation (Thm 7.1)	B	RCF_n §7.1.4	P2 — decoherence internal; $B_{cross}/B_{intra} \to \infty \Rightarrow$ off-diagonal correlations vanish
7.4	Redundant encoding (Def 7.3)	B	RCF_n §7.2.2	$\mathcal{R}_{\eta}(R) = \#\{E_k : Corr_{\omega}(E_k, R) \geq \eta\}$
7.5	m-Robust record (Def 7.4)	B	RCF_n §7.2.3	Disturbing ≤ m fragments leaves ≥ 1 recoverable fragment
7.6	Redundant Record Robustness (Thm 7.2)	B	RCF_n §7.2.4	P3 — $N_R > m \Rightarrow R$ is m-robust; objectivity = structural redundancy
7.7	Positive Effect Family (Def 7.5)	A	RCF_n §7.3.2	$\{E_i\}$ positive operators with $\Sigma_i E_i = 1_{phys}$
7.8	Record Weights $p_i = \omega(E_i)$ (Def 7.6)	A	RCF_n §7.3.3	Branch weights — probabilistic, Layer A
7.9	Normalization and Positivity (Thm 7.3)	A	RCF_n §7.3.4	P4 — $p_i \geq 0$, $\Sigma_i p_i = 1$; formal probability structure closed
7.10	Born Weights as Sectorwise Zero-Decomp (Thm 7.4)	A→B	RCF_n §7.4	P5 — $\mathcal{M}_B = \Sigma_i p_i C_i^{\dagger} C_i$; $\omega(\mathcal{M}_B) = 0$
7.11	Zero-Envariance (Def 7.7)	A→B	RCF_n §7.5.1 + Sec_7_2 §7.2	MOE fixed-point symmetry under SOE spectral-label swapping — DERIVED from RCF primitives
7.12	Phase Invariance (Lemma 7.1)	A→B	RCF_n §7.5.2	Z-envariance of phase rotations $\Rightarrow p_i$ depends only on $\|c_i\|$

#	Structural Unit	Layer	Source	Status / Notes
7.13	Equal-Magnitude Branch Equality (Lemma 7.2)	A→B	RCF_n §7.5.3	Z-envariance of equal-magnitude swaps ⇒ $p_1 = p_2$ when $|c_1| = |c_2|$
7.14	The Born Rule (Thm 7.5)	A→B	RCF_n §7.5.4 + Sec_7_2 §7.2	P6 — $p_i = |c_i|^2$; T-4 STRENGTHENED; Z-envariance derived as MOE fixed-point
7.15	Measurement as MOE Reconciliation (Conj 7.4.1)	A→B	Sec_7_2 §7.4	P7 — structural conjecture; measurement = MOE descent event within ker($\hat{M}_\omega$)
7.16	Collapse as Fast MOE Descent (Conj 7.4.2)	A→B	Sec_7_2 §7.4	P7 — structural conjecture; Q-10 quarantine maintained
7.17	Nine Consolidated Guardrails	A→B	RCF_n §7.6.1 + Sec_7_2 §7.3	P7 — including FIREWALL (guardrail 7); no conflation of probability and gravity

Table 7.7.1 — Seventeen structural units of Section 7, by layer, source, and status.

The cumulative architectural chain of Section 7 is:

Conceptual chain of Section 7

```
master constraint = 0  (§1.1)

        |

        ▼

sector-weighted zero support □_B = Σ_i p_i C_i† C_i  (Thm 7.4, P5)

        |

        ▼

stable record sectors □_i^ω ⊂ ker(M̂_ω)  (Def 7.1)

        |   with B_cross(i,j) ≫ B_intra  (Thm 7.1, P2)

        ▼

redundant m-robust encoding □_η(R) > m  (Thm 7.2, P3)

        |

        ▼

normalized record weights p_i = ω(E_i)  (Thm 7.3, P4)

        |   with p_i ≥ 0, Σ p_i = 1
```

```
        ▼
Z-envariance as MOE fixed-point symmetry  (Def 7.7, P6)
        |   phase invariance (Lem 7.1) + equal-magnitude equality (Lem 7.2)
        ▼
Born rule p_i = |c_i|²  (Thm 7.5, T-4 STRENGTHENED)
        |
        ▼
operational probabilities P(α) = ⟨ψ|Π̂_α|ψ⟩
        |
        ▼
measurement as MOE descent (Conj 7.4.1, P7 — structural conjecture)
        |
        ▼
collapse as fast MOE descent (Conj 7.4.2, P7 — Q-10 maintained)
```

This completes the probability and classicality layer. The framework now possesses an emergent spacetime, populated by interacting matter, governed by gravity, capable of explaining definite observational outcomes, and protected by the FIREWALL guardrail that keeps probability (Layer A) structurally distinct from curvature (Layer C). The final step is to apply this relational architecture to the universe as a whole — Section 8 (Cosmology).